AI Agents for Astronomical Research

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BOOKS Brief it like a new Master's student

B **Bounds** — what must it not do?

```
never modify raw snapshots; Python + numpy/matplotlib; ask before steps > 10 min
```

O **Objective** — what should it achieve?

```
measure how the 10/50/90% Lagrangian radii evolve; find core collapse
```

O **Output** — what comes back?

```
lagrange.py + one labelled figure + 5-line note
```

K **Knowns** — what must it know?

```
snapshots in run01/, N-body (Hénon) units, scales in the header
```

S **Success** — how will we judge it's good?

```
radii at t=0 match the IC file within 1%; report Myr and pc
```

Objective = what to achieve · Output = what you get back · Success = how we judge it

GUARD Second round — when the first answer isn't good enough

G **Grounding / tools** — point it at the real sources and tools (ADS, your files, your code) and ask it to cite where each claim comes from

U **Uncertainty** — allow "I don't know"; ask it to flag what it is unsure about

A **Audience / role** — who the result is for, and at what level

R **Reference examples** — if it keeps getting it wrong, show it one good example (an iteration step, not a first-round must)

D **Done checks** — say how it must check its own work before calling it done (tests, a sanity plot, a known limit)

Verify **Let the agent act on reversible steps; keep a human gate on irreversible or science-conclusion steps.**

Checks: **git diff** · **tests** · **independent reference calculation** · **convergence check** · **known limits** · **physical sanity**

Resources

`code.claude.com/docs/en/best-practices`
`learn.chatgpt.com/docs/codex/cli`
Going further: `github.com/mattpocock/skills`
`docs.nersc.gov/development/coding-agents`
`github.com/adsabs/scix-mcp`
Basic AI Skills (Moodle): `moodle.uni-heidelberg.de/course/view.php?id=27651`

Data note

Heidelberg University guidance: don't enter sensitive, personal or confidential data, or copyrighted text, into AI systems that process data outside university infrastructure.

Your turn

1 One task you repeat in your research:

_____

_____

2 Split it into 3–6 steps; mark each: agent / human check

Step	Describe the step	agent	human check
1		☐	☐
2		☐	☐
3		☐	☐
4		☐	☐
5		☐	☐
6		☐	☐

For one agent step: **Success** — how I'll judge it: _____

Output — what comes back: _____

3 Now brief it with BOOKS

B **Bounds**

O **Objective**

O **Output**

K **Knowns**

S **Success**

Agent = Model + Harness + Context — context is the part you control.